Machine Learning Classification of VMO- and VL-Dominant Muscle Activation Patterns from Surface Electromyography During Squatting

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***Abstract*—Patellofemoral pain and other knee-related movement disorders have been associated with altered activation patterns of the vastus medialis obliquus (VMO) and vastus lateralis (VL), motivating objective methods for characterizing quadriceps muscle recruitment. Surface electromyography (sEMG) provides a non-invasive means of measuring muscle activation, while machine-learning methods can identify patterns within multivariate signal features. This study investigates how effectively machine-learning models can classify VMO- and VL-dominant muscle activation patterns during squatting and determines which evaluated classifier achieves the highest performance. A publicly available squatting sEMG dataset was processed into 998 observations comprising 157 VMO-dominant and 841 VL-dominant observations. Five features were used for classification: VMO root-mean-square amplitude, VL root-mean-square amplitude, peak activation delay, muscle onset delay, and inter-channel synchronization measured through correlation. Three supervised classifiers—Random Forest, Support Vector Machine (SVM), and Decision Tree—were evaluated using a stratified 70:30 train-test split. Class imbalance was addressed through class weighting, and performance was assessed using accuracy, balanced accuracy, precision, recall, and F1-score. Random Forest achieved an accuracy of 96.67% and balanced accuracy of 92.83%. SVM achieved 96.33% accuracy and the highest balanced accuracy of 97.83%, while Decision Tree achieved 89.33% accuracy and 87.61% balanced accuracy. Random Forest produced the highest overall accuracy, whereas SVM demonstrated the strongest class-balanced performance. The results demonstrate that combining amplitude, temporal, and inter-muscular synchronization features can support effective classification of VMO- and VL-dominant activation patterns, while also highlighting the importance of class-aware evaluation and further validation before clinical application.**

***Index Terms*—surface electromyography, sEMG, VMO, vastus medialis obliquus, vastus lateralis, machine learning, Random Forest, Support Vector Machine, rehabilitation**

I. Introduction

Patellofemoral pain is a common musculoskeletal condition in which altered biomechanics and quadriceps muscle recruitment may contribute to abnormal patellar tracking. The vastus medialis obliquus (VMO) and vastus lateralis (VL) are two components of the quadriceps muscle group that exert opposing medial and lateral influences on the patella. Previous work has investigated differences in quadriceps activity between individuals with and without patellofemoral pain, motivating the use of objective measurements of muscle activation rather than relying exclusively on visual or subjective assessment [1].

Surface electromyography (sEMG) provides a non-invasive method for recording electrical activity associated with muscle contraction. The resulting signals contain information about activation magnitude, timing, and relationships between muscles. However, raw EMG signals are oscillatory and noisy, making direct interpretation difficult. Signal processing and feature extraction are therefore required before machine-learning methods can be applied.

Machine learning has increasingly been applied to sEMG classification and rehabilitation engineering. Support Vector Machines have been used to classify sEMG signals for upper-limb self-rehabilitation training using time-domain features such as root-mean-square amplitude [3]. Tree-based and ensemble learning methods have likewise been applied to EMG-based movement classification. Lower-limb applications have demonstrated that sEMG can contain sufficient information to distinguish movement or pathology-related classes [4]. In particular, temporal characteristics of VMO and VL activation have also been investigated in classification of patellofemoral osteoarthritis [5].

These findings suggest that VMO and VL activation should not be characterized solely by their absolute amplitudes. Relative timing and inter-muscular coordination may also provide discriminative information. This motivates the development of a multifeature classification pipeline incorporating both magnitude- and timing-based descriptors.

The present study uses a publicly available squatting sEMG dataset released by Mohr [2]. The data were processed into window-level observations and classified into two categories: VMO-dominant and VL-dominant activation. Three supervised classifiers with different decision mechanisms were evaluated: Random Forest, Support Vector Machine (SVM), and Decision Tree.

The research question addressed by this work is:

> *How effectively can machine-learning models classify VMO- and VL-dominant muscle activation patterns from surface electromyography (EMG) features during squatting, and which evaluated model achieves the highest classification performance?*

The objective was therefore not merely to obtain a high classification score, but to compare multiple classifiers under the same data-processing framework and determine the most effective model according to quantitative evaluation metrics.

II. Background and Related Work

A. VMO–VL Activation and Patellofemoral Mechanics

The VMO and VL contribute to quadriceps force generation and influence patellar motion. Differences in activation between these muscles have been investigated in relation to patellofemoral pain and altered knee mechanics. Boucher et al. examined quadriceps femoris muscle activity in individuals with patellofemoral pain, establishing a physiological basis for examining quadriceps activation patterns in this context [1].

A useful conceptual measure of muscle balance is the relative activation of the VMO and VL. A lower VMO-to-VL activation ratio indicates greater relative VL activation, whereas a higher ratio indicates greater relative VMO activation. However, classification based on a single ratio can discard information contained in activation timing and coordination.

B. sEMG-Based Machine Learning

sEMG has become an important signal source for human movement classification because it provides information about muscle activation without requiring invasive measurements. Cai et al. demonstrated the use of an SVM-based sEMG classification system for upper-limb self-rehabilitation training, using conventional time-domain descriptors including RMS-related measures [3].

Naik et al. applied an ICA-based energy method to sEMG signals for lower-limb movement classification in individuals with and without knee pathology, demonstrating the feasibility of extracting discriminative information from lower-limb EMG signals [4]. Ibraheem et al. further demonstrated the value of timing information from VMO and VL signals for classification involving patellofemoral osteoarthritis [5].

These studies motivate the use of multiple feature domains in the present work. RMS amplitude captures activation magnitude, while peak and onset timing capture temporal coordination. Correlation between the two channels provides an additional measure of their synchronized activation.

C. Machine Learning for Rehabilitation

Different machine-learning algorithms impose different assumptions on the relationship between input features and class labels. Decision Trees provide interpretable hierarchical decision boundaries, while Random Forests combine multiple decision trees to reduce variance and improve robustness. SVMs instead identify a separating boundary in feature space and can model nonlinear relationships through kernels.

Prior sEMG rehabilitation research has demonstrated successful applications of both SVM and Random Forest approaches [3], [6]. Comparing models under identical preprocessing therefore provides an empirical basis for selecting an appropriate classifier for the present classification task.

III. Methodology

A. Dataset and Data Provenance

The analysis used a publicly available dataset entitled *Data For: Intermuscular Coherence between Surface EMG Signals Recorded with Monopolar and Bipolar EMG Systems during Squatting*, released by Mohr through Mendeley Data in 2018 [2]. The dataset contains EMG recordings associated with squatting and provides the underlying two-channel signals used for the present secondary analysis.

No new human-subject data were collected for this study. All preprocessing, feature construction, classification, and evaluation were performed computationally on the existing dataset.

The exact sampling frequency, acquisition hardware specifications, and some electrode-level acquisition parameters were not retained in the project files available for the present analysis. These parameters are therefore not inferred in this manuscript.

B. Channel Identification

The two recorded channels were interpreted as VMO and VL. The project analysis identified Channel 1 as VMO and Channel 2 as VL based on the relative activation magnitude and timing observed in the processed signals. Channel 2 exhibited earlier activation and somewhat greater average activity, which was consistent with the project's identification of that channel as VL.

This identification was subsequently used consistently throughout feature extraction and classification.

C. Signal Processing and Windowing

The raw EMG signals were transformed into window-level observations so that local activation patterns could be represented numerically.

The processing workflow consisted of:

1) loading the two-channel EMG data;
2) removing indexing and positional metadata;
3) processing the two muscle channels;
4) calculating activation features for each observation;
5) assigning each observation to a VMO-dominant or VL-dominant class; and
6) training and evaluating supervised classification models.

The processed dataset contained 998 observations. The class distribution was:

- VMO-dominant: 157 observations (15.7%);
- VL-dominant: 841 observations (84.3%).

The resulting distribution is substantially imbalanced. A classifier predicting the majority VL-dominant class for every observation would already obtain approximately 84.3% accuracy. Consequently, accuracy alone is insufficient for evaluating the minority class.

D. Feature Construction

The original processed dataset contained seven candidate features, including RMS and integrated-EMG ratios. However, the final Random Forest, SVM, and Decision Tree notebooks explicitly removed both `RMS_ratio` and `iEMG_ratio` before training.

Consequently, the final classifiers were trained using five features:

TABLE I
FEATURES USED BY THE FINAL MACHINE-LEARNING CLASSIFIERS

Feature	Symbol	Description
VMO RMS	R_{VMO}	VMO activation magnitude
VL RMS	R_{VL}	VL activation magnitude
Peak delay	Δt_{peak}	Difference in peak activation time
Onset delay	$\Delta t_{\mathrm{onset}}$	Difference in activation onset
Synchronization	ρ	Inter-channel correlation

1) Root-Mean-Square Activation: RMS was used to quantify the magnitude of activation within each channel. For a window containing N signal samples x_i, RMS is defined as:

$$RMS = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2}. \tag{1}$$

The two RMS values therefore represented the activation magnitude of the VMO and VL channels independently.

2) Peak Activation Delay: Peak activation time was determined by identifying the index corresponding to the maximum signal amplitude:

$$t_{\mathrm{peak}} = \arg\max_t |x(t)|. \tag{2}$$

The relative peak delay between the channels was then calculated as:

$$\Delta t_{\mathrm{peak}} = t_{\mathrm{peak,VL}} - t_{\mathrm{peak,VMO}}. \tag{3}$$

Positive values indicate that the VL peak occurred later than the VMO peak under this definition, while negative values indicate earlier VL activation.

3) Activation Onset Delay: The raw-input prediction implementation used a threshold-based approach for identifying activation onset. The threshold was defined as 10% of the maximum rectified signal amplitude:

$$T = 0.1 \max(|x(t)|). \tag{4}$$

The onset was identified from the first sample exceeding the threshold. The difference between VL and VMO onset was then calculated as:

$$\Delta t_{\mathrm{onset}} = t_{\mathrm{onset,VL}} - t_{\mathrm{onset,VMO}}. \tag{5}$$

This feature represents relative activation timing rather than absolute activation duration.

4) Inter-Muscular Synchronization: The linear relationship between the two muscle channels within each window was quantified using Pearson correlation:

$$\rho = \frac{\sum_{i=1}^{N}(v_i - \bar{v})(l_i - \bar{l})}{\sqrt{\sum_{i=1}^{N}(v_i - \bar{v})^2 \sum_{i=1}^{N}(l_i - \bar{l})^2}}. \tag{6}$$

Here, v_i and l_i represent the VMO and VL signal values, respectively. Higher values indicate stronger simultaneous variation between the channels.

E. Data Cleaning

The model notebooks removed non-predictive indexing variables including window identifiers and start/end indices. The candidate ratio features were also removed from the final model feature matrix.

Missing values were handled using mean imputation:

$$x_{\mathrm{missing}} = \frac{1}{n}\sum_{i=1}^{n} x_i. \tag{7}$$

The project data contained no missing values after the processed dataset was loaded, but the imputation step was retained as part of the modelling pipeline.

F. Train-Test Split

The dataset was divided into training and testing subsets using a 70:30 split. Stratification was applied so that the relative class proportions were maintained across the two subsets.

A fixed random state of 42 was used:

```
train_test_split(
    X, y,
    test_size=0.3,
    random_state=42,
    stratify=y
)
```

The resulting test set contained 300 observations:

- 47 VMO-dominant observations;
- 253 VL-dominant observations.

G. Feature Scaling

Standardization was performed using the training data:

$$z = \frac{x - \mu}{\sigma}, \tag{8}$$

where μ and σ were calculated from the training set.

The same transformation was then applied to the held-out test set. This prevented the test observations from influencing the scaling parameters.

IV. MACHINE-LEARNING MODELS

Three supervised classification algorithms were evaluated.

A. Random Forest

The Random Forest classifier consisted of 200 decision trees. A fixed random state of 42 was used, and class weighting was set to `balanced_subsample` to address the unequal class distribution.

The principal configuration was:

```
RandomForestClassifier(
    n_estimators=200,
    random_state=42,
```

```
    class_weight="balanced_subsample"
)
```

Random Forest was selected because its ensemble structure can capture nonlinear interactions between signal features while reducing the variance associated with an individual decision tree.

B. Support Vector Machine

The SVM used a radial basis function (RBF) kernel. The hyperparameters were:

```
SVC(
    kernel="rbf",
    C=100,
    gamma=0.01,
    class_weight="balanced",
    probability=True
)
```

The RBF kernel permits nonlinear separation of observations in the standardized feature space. Balanced class weights were used to prevent the majority VL-dominant class from dominating the optimization.

C. Decision Tree

The Decision Tree classifier was configured with a maximum depth of five:

```
DecisionTreeClassifier(
    max_depth=5,
    class_weight="balanced",
    random_state=42
)
```

The depth restriction limits tree complexity and reduces the risk of fitting highly specific training observations.

V. Evaluation Metrics

Performance was evaluated using accuracy, balanced accuracy, precision, recall, and F1-score.

A. Accuracy

Accuracy measures the proportion of all test observations classified correctly:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}. \tag{9}$$

Although useful as an overall measure, accuracy can be misleading for imbalanced datasets.

B. Balanced Accuracy

Balanced accuracy is the arithmetic mean of sensitivity for the two classes:

$$Balanced\ Accuracy = \frac{Recall_{\mathrm{VMO}} + Recall_{\mathrm{VL}}}{2}. \tag{10}$$

This metric gives equal importance to the two classes regardless of their frequency.

C. Precision and Recall

Precision measures the proportion of predicted positives that are correct:

$$Precision = \frac{TP}{TP + FP}. \tag{11}$$

Recall measures the proportion of actual positives that are correctly identified:

$$Recall = \frac{TP}{TP + FN}. \tag{12}$$

Both metrics were calculated independently for VMO-dominant and VL-dominant observations.

D. F1-Score

The F1-score is the harmonic mean of precision and recall:

$$F1 = 2\frac{Precision \cdot Recall}{Precision + Recall}. \tag{13}$$

F1-score is particularly informative for the minority VMO-dominant class because it penalizes a model that achieves high recall at the expense of precision, or vice versa.

VI. Results

A. Overall Model Performance

All three classifiers achieved substantially greater accuracy than the approximately 84.3% majority-class baseline.

TABLE II
Overall Performance of the Three Evaluated Classifiers

Model	Accuracy	Balanced Accuracy
Random Forest	**96.67%**	92.83%
SVM	96.33%	**97.83%**
Decision Tree	89.33%	87.61%

Random Forest achieved the highest overall accuracy at 96.67%, narrowly exceeding the SVM accuracy of 96.33%. The Decision Tree achieved substantially lower accuracy at 89.33%.

SVM, however, achieved the highest balanced accuracy at 97.83%. This distinction is important because the research objective involves classification of both VMO- and VL-dominant activation patterns rather than simply maximizing the number of correctly classified observations.

B. Per-Class Performance

The complete per-class results are shown in Table III.

The models displayed different performance profiles.

Random Forest correctly identified 87% of VMO-dominant test observations and 98% of VL-dominant observations. Its VMO-dominant F1-score was 0.89.

SVM identified all VMO-dominant observations in the test set, producing 100% recall for the minority class. However, its VMO precision was only 81%, indicating that some observations predicted as VMO-dominant were actually VL-dominant. The resulting VMO F1-score was 0.90.

TABLE III
PER-CLASS PRECISION, RECALL, AND F1-SCORE FOR THE EVALUATED CLASSIFIERS

	VMO-Dominant			VL-Dominant		
Model	Precision	Recall	F1	Precision	Recall	F1
Random Forest	0.91	0.87	0.89	0.98	0.98	0.98
SVM	0.81	1.00	0.90	1.00	0.96	0.98
Decision Tree	0.62	0.85	0.71	0.97	0.90	0.93

Decision Tree produced the weakest minority-class precision at 62% and the lowest VMO F1-score at 0.71. Its VL-dominant recall remained relatively high at 90%, reflecting the model's stronger performance on the majority class.

C. Confusion Matrices

The final model notebooks preserved the following confusion matrices.

TABLE IV
RANDOM FOREST CONFUSION MATRIX

	Predicted VMO	Predicted VL
Actual VMO	41	6
Actual VL	4	249

TABLE V
SUPPORT VECTOR MACHINE CONFUSION MATRIX

	Predicted VMO	Predicted VL
Actual VMO	47	0
Actual VL	11	242

TABLE VI
DECISION TREE CONFUSION MATRIX

	Predicted VMO	Predicted VL
Actual VMO	40	7
Actual VL	25	228

The Random Forest matrix shows a relatively small number of errors for both classes. Six VMO-dominant observations were classified as VL-dominant, while four VL-dominant observations were classified as VMO-dominant.

The SVM matrix demonstrates its high minority-class recall: all 47 VMO-dominant observations were identified correctly. However, 11 VL-dominant observations were incorrectly classified as VMO-dominant.

The Decision Tree produced substantially more VL-to-VMO misclassifications, with 25 VL-dominant observations incorrectly classified as VMO-dominant.

D. Model Selection

The answer to the research question depends on the performance criterion used.

Random Forest achieved the highest accuracy:

$$96.67\% > 96.33\% > 89.33\%. \tag{14}$$

SVM achieved the highest balanced accuracy:

$$97.83\% > 92.83\% > 87.61\%. \tag{15}$$

Therefore, if "highest classification performance" is defined primarily by overall accuracy, Random Forest is the best-performing classifier. If equal performance across the two classes is prioritized, SVM provides the stronger result.

Because the primary model-selection objective of this project was to determine the classifier with the highest overall classification accuracy while also examining class-aware metrics, Random Forest was selected as the overall best-performing model.

VII. DISCUSSION

A. Interpretation of Model Performance

The results demonstrate that machine-learning classifiers can distinguish VMO- and VL-dominant activation patterns from a relatively small set of sEMG-derived features.

The strongest overall accuracy was obtained by Random Forest at 96.67%. SVM produced a nearly identical accuracy of 96.33%, while Decision Tree was substantially weaker at 89.33%.

The close performance of Random Forest and SVM suggests that the five-dimensional feature representation contains substantial class-discriminative information. Both models can capture nonlinear relationships that are difficult to represent using a single threshold or simple ratio.

The weaker performance of the Decision Tree suggests that a single hierarchical decision structure may be less effective for this feature space. Although the maximum tree depth was limited to five to constrain complexity, the model may still have been less capable of representing the distributed relationships between activation magnitude, timing, and synchronization.

B. Importance of Class Imbalance

The dataset contained 841 VL-dominant observations compared with only 157 VMO-dominant observations. This imbalance is consequential.

A classifier that simply predicted VL dominance for every observation would obtain approximately 84.3% accuracy without learning any meaningful distinction. The fact that all three models exceeded this baseline is therefore informative, but it does not by itself establish balanced performance.

The balanced accuracy results reveal a different picture from ordinary accuracy. SVM achieved the highest balanced accuracy at 97.83%, substantially exceeding its ordinary accuracy by demonstrating excellent sensitivity across both classes. Random Forest achieved 92.83% balanced accuracy, while Decision Tree achieved 87.61%.

The use of class weighting in all three classifiers was therefore important. Random Forest used `balanced_subsample`, while SVM and Decision Tree used `balanced` class weights.

C. Minority-Class Behavior

The minority VMO-dominant class provides an especially useful comparison.

SVM achieved 100% recall for VMO-dominant observations, meaning that none of the 47 VMO-dominant test observations were missed. However, this came with a precision of only 81%, because 11 VL-dominant observations were also classified as VMO-dominant.

Random Forest produced a more conservative classification pattern. Its VMO recall was 87%, but its precision was 91%. Consequently, the F1-score of 0.89 was close to the SVM value of 0.90.

This distinction demonstrates why relying on a single metric can produce an incomplete evaluation. SVM was better at ensuring that VMO-dominant observations were not missed, whereas Random Forest produced a more balanced trade-off between false positives and false negatives.

D. Role of the Feature Set

The final classifiers used five features spanning three information domains.

First, VMO and VL RMS values represented activation magnitude. Second, peak and onset delays represented temporal coordination. Third, inter-channel correlation represented synchronization.

This combination is important because two observations can have similar overall activation magnitude while differing substantially in timing. Similarly, two muscles can activate at similar times while exhibiting different amplitudes or degrees of synchronization.

The project initially generated RMS and integrated-EMG ratios as candidate features. However, these two ratio features were explicitly removed in the final model notebooks. The final results should therefore be interpreted as evidence for classification using the five retained features rather than as evidence that the VMO/VL RMS ratio itself was responsible for the model performance.

E. Relationship to Previous Research

The findings are consistent with previous work demonstrating that sEMG features can support machine-learning classification in rehabilitation contexts. Cai et al. showed the feasibility of SVM-based classification using sEMG-derived time-domain features [3]. Naik et al. demonstrated lower-limb sEMG classification in individuals with and without knee pathology [4]. Ibraheem et al. specifically demonstrated the relevance of VMO/VL timing information to classification involving patellofemoral osteoarthritis [5].

The present study differs in its specific classification objective. Rather than directly predicting the presence of a disease, it classifies activation dominance between two quadriceps muscles. This distinction is important because the current dataset and experiment do not provide sufficient evidence to claim that the model diagnoses patellofemoral pain or any other clinical condition.

VIII. Potential Rehabilitation Application

The classification framework may be relevant to rehabilitation monitoring because muscle activation patterns can provide information about how an individual recruits the quadriceps during movement.

A potential application would involve acquiring two-channel sEMG during a standardized exercise, extracting the same five features, and passing them through the trained classifier. The resulting prediction could identify whether the observed activation window is more consistent with VMO-dominant or VL-dominant recruitment.

Such a system could potentially provide objective feedback during rehabilitation exercises. However, the present experiment does not establish that a particular predicted class corresponds directly to clinical improvement or deterioration.

The system should therefore be regarded as a computational classification framework rather than a validated clinical decision-support tool.

IX. Limitations

Several limitations constrain the interpretation of the results.

A. Dataset Imbalance

The processed dataset contains a strong 84.3:15.7 imbalance between VL-dominant and VMO-dominant observations. Class weighting and balanced accuracy partially address this issue, but the minority class remains substantially smaller.

B. Window-Level Validation

The observations are derived from windowed EMG signals. Adjacent windows may contain related or overlapping signal information. A random window-level train-test split can therefore produce a more optimistic estimate of generalization than a subject-wise or trial-wise split if windows from the same underlying recording are represented in both subsets.

A future study should use subject-level or trial-level separation where the original dataset metadata permit it.

C. Limited Dataset Diversity

All observations originate from one publicly available dataset. Consequently, the reported performance does not establish generalization to other individuals, acquisition systems, exercise protocols, electrode placements, or clinical populations.

External validation on an independent dataset is necessary before clinical claims can be made.

D. Acquisition Metadata

The project files do not preserve all original acquisition metadata, including the exact sampling frequency and complete hardware configuration. These parameters should be documented explicitly in a future reproducibility release.

E. Onset Detection

The raw-input prediction pipeline identifies onset using a threshold equal to 10% of maximum signal amplitude. Although computationally simple and consistent, this approach is less robust than methods based on a statistically estimated baseline and sustained threshold crossing.

The project documentation considered alternative baseline-based approaches, including estimating the mean and standard deviation of a low-activity region and defining a threshold based on baseline statistics. These approaches should be evaluated in future work.

F. Single Train-Test Split

The reported results originate from one stratified 70:30 split using random state 42. Although the fixed split makes the experiment reproducible, a single split does not provide an estimate of performance variability.

Repeated stratified cross-validation would provide more robust estimates and confidence intervals.

X. Reproducibility and Data Availability

The dataset used in this study is publicly available through the Mendeley Data repository described by Mohr [2].

The analysis was implemented in Python using standard scientific-computing and machine-learning libraries, including Pandas, NumPy, Matplotlib, Seaborn, and scikit-learn.

The final model notebooks document the preprocessing and model configurations used to obtain the reported results. In particular, all three classifiers use the same five-feature representation, the same 70:30 stratified split, and the same random state of 42. The SVM hyperparameters and the principal Random Forest and Decision Tree configurations are explicitly specified in the corresponding notebooks.

For full reproducibility, future versions of the project should additionally preserve the exact processed dataset, acquisition metadata, preprocessing parameters, and complete model outputs alongside the source code.

XI. Future Work

Several extensions would strengthen the present work.

First, the classification pipeline should be evaluated using repeated stratified cross-validation rather than a single train-test split. This would allow model performance to be reported with measures of uncertainty.

Second, subject-wise or trial-wise validation should be investigated to determine whether the models generalize beyond highly related signal windows.

Third, external validation using an independent sEMG dataset would determine whether the extracted features generalize across participants and acquisition conditions.

Fourth, more robust EMG onset-detection methods should be compared with the current percentage-of-maximum threshold. Baseline-dependent methods could reduce sensitivity to noise and variation in signal amplitude.

Fifth, explainability methods could be used to determine which features contribute most strongly to the classification decisions. The Random Forest implementation calculates feature importance, while the SVM implementation uses permutation importance. A unified analysis of these importance measures could identify whether magnitude, timing, or synchronization features provide the strongest discriminative information.

Finally, the trained model could be integrated into a real-time rehabilitation-monitoring system. Such a system would require standardized electrode placement, calibrated signal acquisition, real-time feature extraction, and clinical validation before being used outside a research setting.

XII. Conclusion

This study developed and evaluated a machine-learning pipeline for classifying VMO- and VL-dominant muscle activation patterns from surface electromyography during squatting.

The final classification pipeline used five features: VMO RMS amplitude, VL RMS amplitude, peak activation delay, onset delay, and inter-channel synchronization. Three supervised classifiers were compared using a stratified 70:30 train-test split: Random Forest, Support Vector Machine, and Decision Tree.

Random Forest achieved the highest overall accuracy at 96.67%, narrowly exceeding SVM at 96.33% and substantially outperforming Decision Tree at 89.33%. Random Forest also achieved 92.83% balanced accuracy. SVM, however, achieved the highest balanced accuracy at 97.83% and obtained 100% recall for the minority VMO-dominant class.

Accordingly, the answer to the research question is that machine-learning models can classify the two activation-dominance classes with high performance within the evaluated dataset, with Random Forest providing the highest overall accuracy and SVM providing the strongest class-balanced performance.

The results support the feasibility of using combined magnitude-, timing-, and synchronization-based sEMG features for automated VMO–VL activation classification. However, the results should not be interpreted as clinical validation or as evidence that the model can diagnose patellofemoral pain. Subject-wise validation, independent datasets, standardized acquisition protocols, and clinical evaluation are required before translation into a rehabilitation decision-support system.

Ethics and Privacy Statement

This study used previously collected, publicly available secondary data and did not involve the collection of new human-subject measurements. The analysis does not attempt to identify individual participants. Any future clinical deployment would require appropriate consideration of informed consent,

data privacy, secure storage of physiological measurements, model fairness across populations, and the risks associated with automated rehabilitation recommendations.

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